Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical
Practical-8

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|---|---|
| Roll No.:T005 | Name: Ritesh chaubey |
| Class:TYIT | Batch:1 |
| Date of Assignment: 23-9-2026 | Date/Time of Submission: 23-9-2026 |

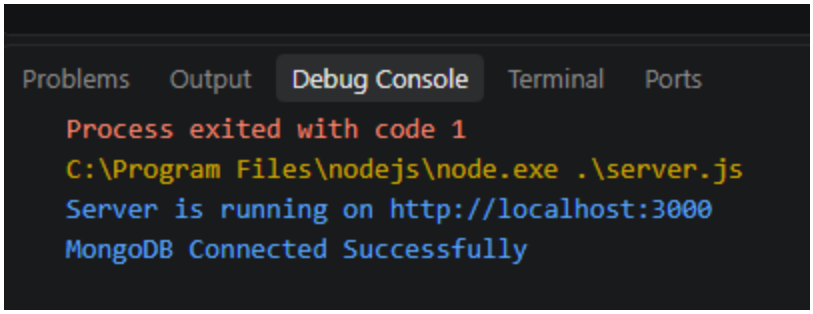
## 8.API Creation and Integration

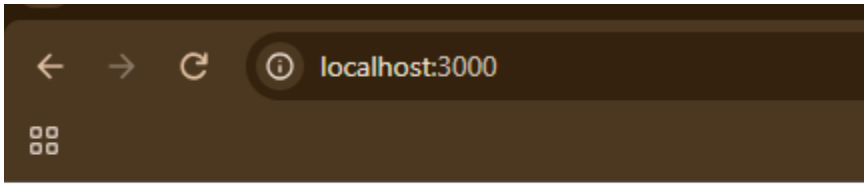
### a. Create a simple REST API using Express

server.js:-

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
.then(() => {
console.log("MongoDB Connected Successfully");
})
.catch((error) => {
console.log("MongoDB Connection Failed");
console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
res.send("Student Management API is Running(Ritesh Chaubey)");
});
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);
});
```

Output:-



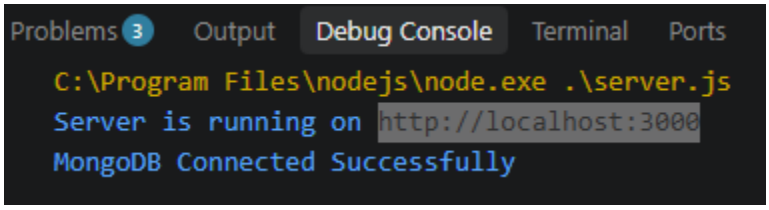

Student Management API is Running(Ritesh Chaubey)

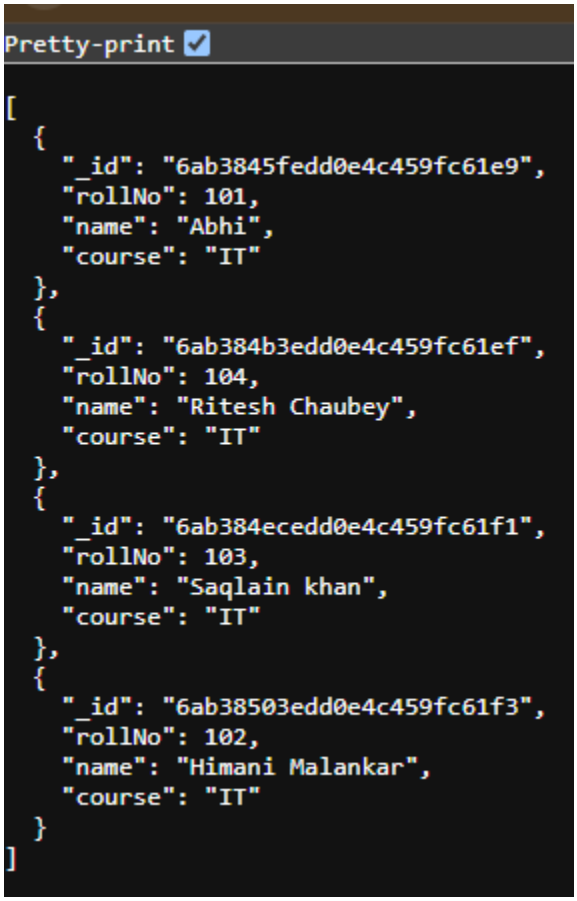
## b. Implement a GET API to retrieve data.

server.js:-

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/college")
.then(() => {
console.log("MongoDB Connected Successfully");
})
.catch((error) => {
console.log("MongoDB Connection Failed");
console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
res.send("Student Management API is Running(Ritesh Chaubey)");
});
app.get("/students", async (req, res) => {
try {
const students = await mongoose.connection.db
.collection("students")
```

```
.find()
.toArray();
res.json(students);
}
catch (error) {
res.status(500).json({
message: "Error fetching students"
});
}
});
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);
});
```



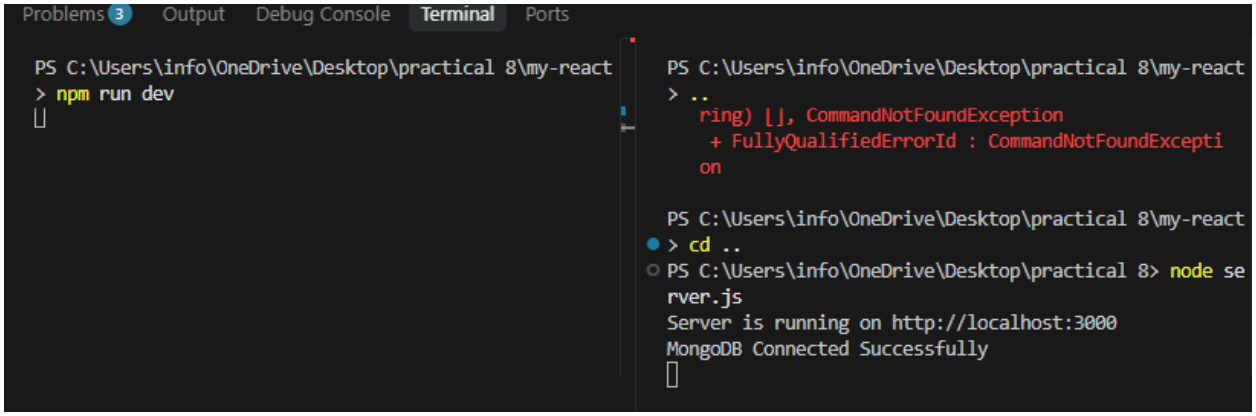

## c. Fetch data from the API in a React application.

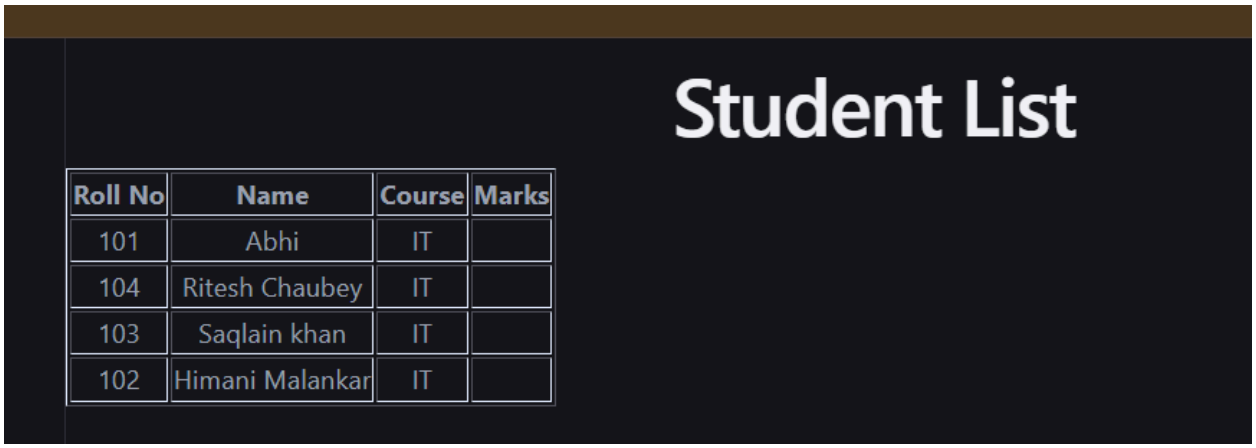
App.jsx:-

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
const [students, setStudents] = useState([]);
useEffect(() => { axios.get("http://localhost:3000/students")
.then((response) => {setStudents(response.data);})

.catch((error) => {console.log(error);}); }, []);
return (
<div>
<h1>Student List</h1>
<table border="1">
<thead>
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Course</th>
<th>Marks</th>
</tr>
</thead>
<tbody>
{students.map((student) => (
<tr key={student._id}>
<td>{student.rollNo}</td>
<td>{student.name}</td>
<td>{student.course}</td>
<td>{student.marks}</td>
</tr>
))}
</tbody>
</table>
</div>
);
}
export default App;
```

Output:-




## D)Send data from React to the server using a POST request.

App.jsx:-

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
const [students, setStudents] = useState([]);
const [rollNo, setRollNo] = useState("");
const [name, setName] = useState("");
const [course, setCourse] = useState("");
const [marks, setMarks] = useState("");
// Get students
const getStudents = () => {axios.get("http://localhost:3000/students")
.then((response) => {setStudents(response.data);})
.catch((error) => {console.log(error);});
};
useEffect(() => {getStudents();}, []);

// Add student
const addStudent = () => {const newStudent = {
rollNo: Number(rollNo),
```

```
name: name,
course: course,
marks: Number(marks)
};
axios.post("http://localhost:3000/students", newStudent)
.then((response) => {alert(response.data.message);
setRollNo("");
setName("");
setCourse("");
setMarks("");
getStudents();
})
.catch((error) => {
console.log(error);
});
};
return (
<div>
<h1>Student Management System</h1>
<h2>Add Student</h2>
<input
type="number"
placeholder="Roll No"
value={rollNo}
onChange={(e) => setRollNo(e.target.value)}
/>
<br /><br />
<input
type="text"
placeholder="Student Name"
value={name}
onChange={(e) => setName(e.target.value)}
/>
<br /><br />

<input
type="text"
placeholder="Course"
value={course}
onChange={(e) => setCourse(e.target.value)}
/>
<br /><br />
<input
type="number"
```

```
placeholder="Marks"
value={marks}
onChange={(e) => setMarks(e.target.value)}
/>
<br /><br />
<button onClick={addStudent}>
Add Student
</button>
<hr />
<h2>Student List</h2>
<table border="1">
<thead>
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Course</th>
<th>Marks</th>
</tr>
</thead>
<tbody>
{students.map((student) => (
<tr key={student._id}>
<td>{student.rollNo}</td>
<td>{student.name}</td>
<td>{student.course}</td>
<td>{student.marks}</td>
</tr>
))}
</tbody>
</table>

</div>
);
}
export default App;
```

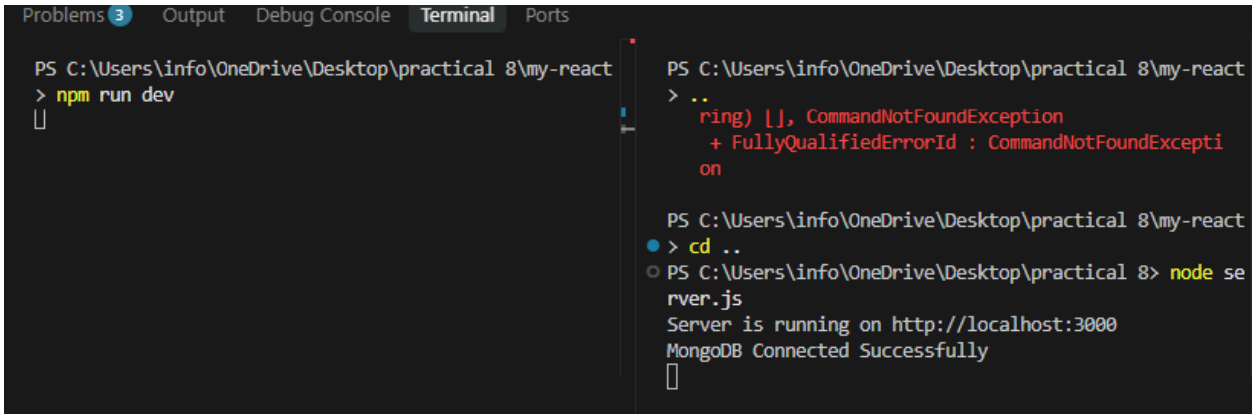
server.js:-

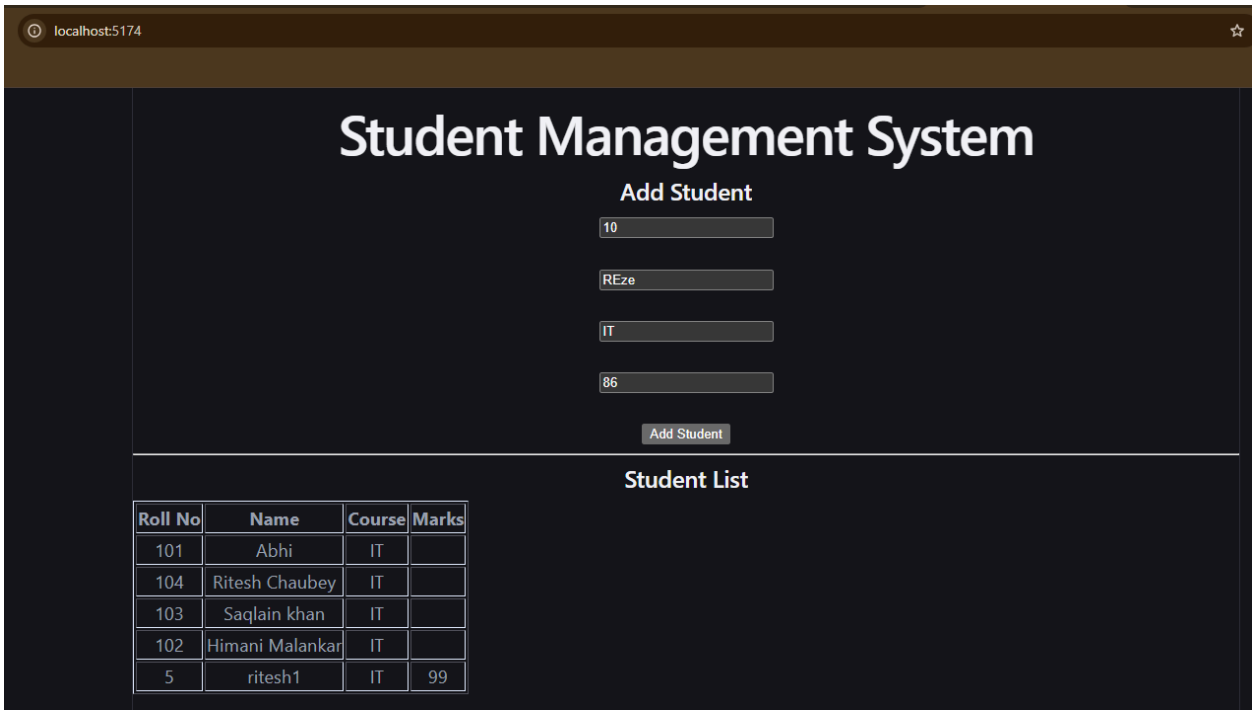
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const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
```

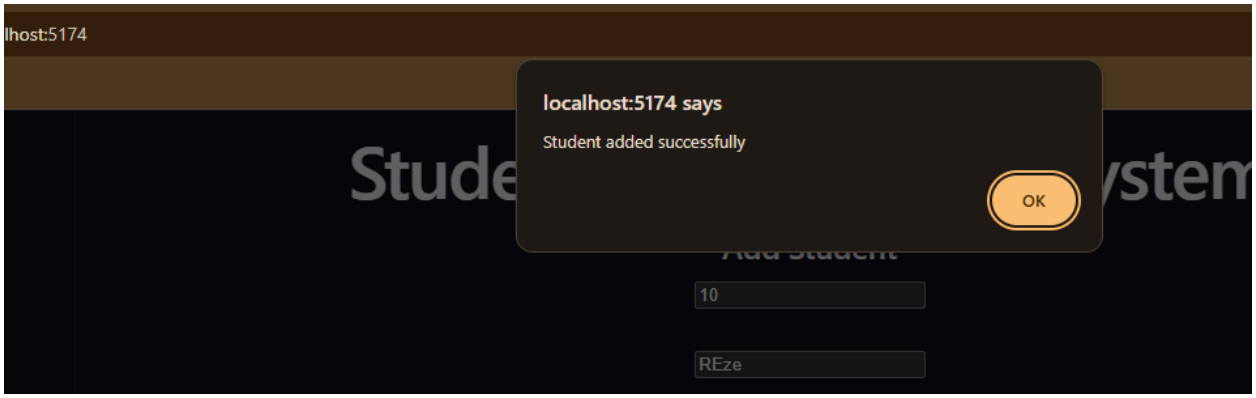
```
app.use(express.json());
// Connect MongoDB
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.then(() => {
console.log("MongoDB Connected Successfully");
})
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console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
res.send("Student Management API is Running(Ritesh Chaubey)");
});
app.get("/students", async (req, res) => {
try {
const students = await mongoose.connection.db
.collection("students")
.find()
.toArray();
res.json(students);
}
catch (error) {
res.status(500).json({
message: "Error fetching students"
});
}
});
app.post("/students", async (req, res) => {
try {
const student = req.body;
await mongoose.connection.db
.collection("students")
.insertOne(student);
res.json({
message: "Student added successfully",
student: student
});
}
catch (error) {
res.status(500).json({
message: "Error adding student"
});
}
```

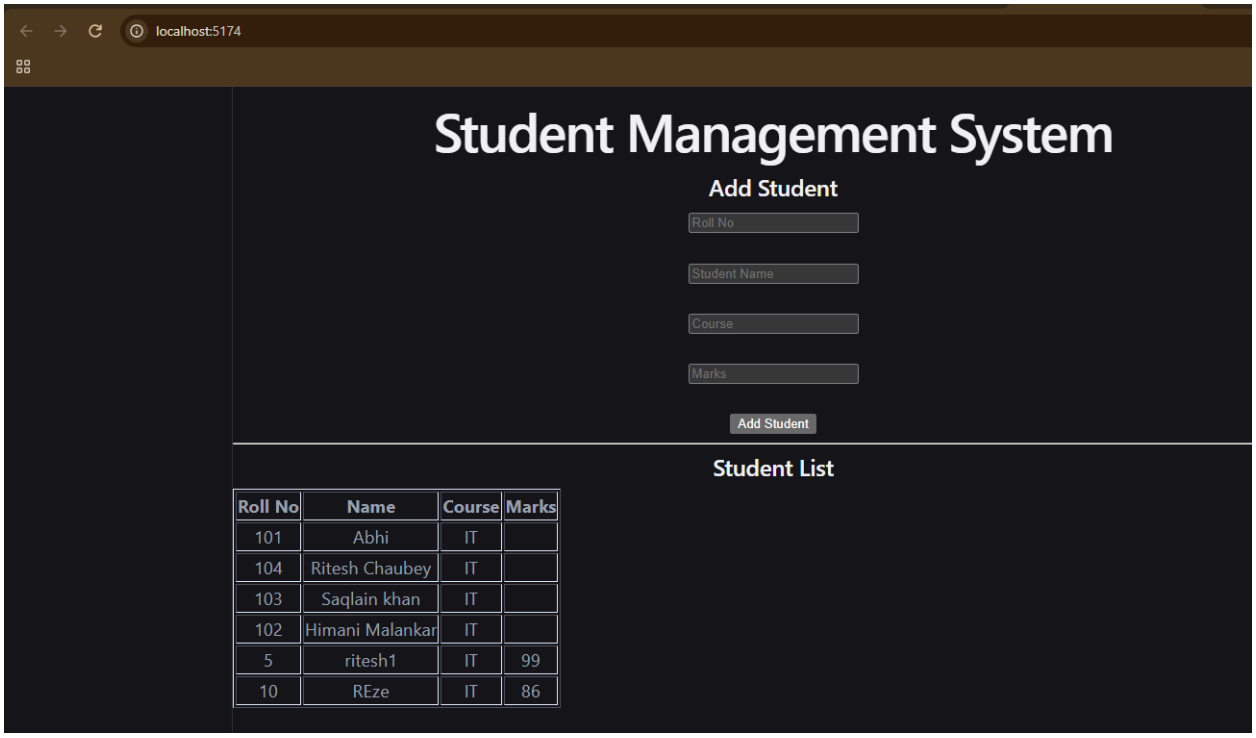
```
});
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);
});
```

Output:-




lhost:5174
localhost:5174 says
Student added successfully
OK
10
REze


localhost:5174
Student Management System
Add Student
Roll No
Student Name
Course
Marks
Add Student
Student List
Roll No | Name | Course | Marks
101 | Abhi | IT |
104 | Ritesh Chaubey | IT |
103 | Saqlain khan | IT |
102 | Himani Malankar | IT |
5 | ritesh1 | IT | 99
10 | REze | IT | 86